A ten-cube counterexample to the recursive structure conjecture for extremal polyominoes

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Abstract

We give a facet-connected union of ten integer unit cubes with bounding box side lengths $(4, 4, 4)$ and convex-hull volume 28, the maximum possible for ten cubes. Every coordinate layer contains at most six cubes. Thus no one-layer subpolyomino can retain both other bounding-box side lengths. This disproves Conjecture 1 of Kurz, already in dimension three.

Task 1: the source question and its resolution

In [1, Conjecture 1, p. 135], an extremal d-dimensional polyomino is conjectured to contain a subpolyomino P' of thickness one in some coordinate direction i, whose projection retains all other bounding-box side lengths and is again extremal; the remainder is to consist of at most two linear arms. The conjecture also asserts that the auxiliary growth parameters v_i vanish. We disprove the existence of P' itself, without needing to interpret the arm condition.

For an integer vector a let $Q_a = a + [0, 1]^3$. Take the following lower corners in the displayed growth order, and set $P = \bigcup_{j=0}^{9} Q_{a_j}$:

j	a_j	earlier facet neighbor
0	$0, 0, 0$	$--$
1	$1, 0, 0$	0
2	$1, 0, -1$	1
3	$1, -1, -1$	2
4	$-1, 0, 0$	0
5	$1, -2, -1$	3
6	$2, 0, 0$	1
7	$1, -1, -2$	3
8	$2, 0, 1$	6
9	$1, -3, -1$	5

(1)

Each listed neighbor differs by one standard basis vector up to sign. Consequently P is a polyomino with ten distinct cubes. Its bounding box is $[-1, 3] \times [-3, 1] \times [-2, 2]$, of side lengths $(4, 4, 4)$. Every added cube increases one bounding-box side length. Thus all the source's parameters v_i are zero. Equivalently, the source identity $n = 1 + \sum_i (l_i - 1) + \sum_i v_i$ forces $\sum_i v_i = 0$.

Theorem 1. *The polyomino P in (1) has the maximum convex-hull volume among ten-cube three-dimensional polyominoes, but has no subpolyomino satisfying conditions (i) and (ii) of Kurz's Conjecture 1.*

Exact convex-hull volume

Write $K = \operatorname{conv}\{a_0, \dots, a_9\}$. Convexity of the unit cube gives

$$\operatorname{conv}(P) = K + [0,1]^3. \tag{2}$$

The six vertices of K are

$$\begin{aligned} A &= (-1,0,0), & B &= (1,0,-1), & C &= (1,-1,-2), \\ D &= (1,-3,-1), & E &= (2,0,0), & F &= (2,0,1). \end{aligned}$$

The other four lower corners lie on the segments AE or BD. Here is a complete facet certificate for this six-vertex polytope. An oriented row UVW contributes $\det(U,V,W)$ to six times the volume. For the quadrilateral use the fan from its first vertex.

Oriented facet	Outward normal u	b in $u \cdot x = b$	Contribution
ACD	$(-5,-2,-4)$	5	5
ADF	$(-3,-5,9)$	3	3
BCA	$(-1,2,-2)$	1	1
$FEBA$	$(0,1,0)$	0	0
CBE	$(1,1,-1)$	2	2
DEF	$(3,-1,0)$	6	6
DCE	$(5,-1,-2)$	10	10

All six vertices satisfy every displayed inequality $u \cdot x \le b$. Checking the $\binom{6}{3} = 20$ vertex triples gives precisely these seven supporting planes; thus the table lists every facet. The origin belongs to K (on AE), so pyramids from the origin to the oriented facets give

$$\operatorname{Vol}(K) = \frac{5+3+1+0+2+6+10}{6} = \frac{9}{2}. \tag{3}$$

The three coordinate projections have the following cyclic vertex lists and areas, directly obtained by the shoelace formula. The letters here denote the corresponding projected points.

Coordinate plane	Projected hull	Area
xy	ADE	9/2
xz	$ACEF$	9/2
yz	$DCBF$	9/2

Every coordinate width of K is 3.

For any three-dimensional convex body L, addition of a unit segment in direction e_i increases its volume by the area of its projection onto $e_i^{\perp}$. Indeed, every nonempty parallel line section is an interval whose length increases by one, and integration over the projection proves the assertion. Applying this three times, together with the two-dimensional analogue for projection areas, gives

$$\begin{aligned} \operatorname{Vol}(K + [0,1]^3) &= \operatorname{Vol}(K) + \operatorname{area}(K_{xy}) + \operatorname{area}(K_{xz}) + \operatorname{area}(K_{yz}) \\ &\quad + \operatorname{width}_x(K) + \operatorname{width}_y(K) + \operatorname{width}_z(K) + 1 \\ &= \frac{9}{2} + 3\frac{9}{2} + 3 + 3 + 3 + 1 = 28. \end{aligned}$$

Kurz's upper bound [1, Theorem 4] for n cubes in dimension d is

$$\sum_{I \subseteq \{1,\dots,d\}} \frac{1}{|I|!} \prod_{i \in I} \left\lfloor \frac{n-2+i}{d} \right\rfloor.$$

For $d = 3$ and $n = 10$ the three floor values are 3, so this equals $1 + 9 + 27/2 + 27/6 = 28$. Hence P is globally extremal for ten cubes.

The missing planar subpolyomino

Lemma 1. *A facet-connected planar polyomino with bounding-box side lengths r and s contains at least $r + s - 1$ cells.*

Proof. Order its cells by growing a spanning tree of the facet-adjacency graph. The first cell has side-length sum two. A subsequent cell shares a side with an earlier cell and can increase that sum by at most one. If there are m cells, then $r + s \leq 2 + (m - 1)$. □

The layer counts of P, indexed by the respective coordinates of its lower corners, are

direction	layer coordinates	numbers of cubes
x	$-1, 0, 1, 2$	$1, 1, 6, 2$
y	$-3, -2, -1, 0$	$1, 1, 2, 6$
z	$-2, -1, 0, 1$	$1, 4, 4, 1.$

In particular, each layer contains at most six cubes. Suppose a subpolyomino P' had thickness one in direction i and retained both other side lengths, as conditions (i) and (ii) require. Its cubes would all lie in one coordinate layer. Projection along i is injective on these cubes and preserves facet connectivity, so it would give a planar polyomino with side lengths $4, 4$ and at most six cells. Lemma 1 requires at least seven. This contradiction proves Theorem 1 and disproves the conjecture. □

Verification and limits

The accompanying standard-library Python verifier uses integer and rational arithmetic. It checks the growth order, layers, all supporting planes of K and of the 44 distinct cube corners, oriented facet volumes, edge incidences, Euler's formula, projection areas, and the source bound. The direct cube-corner hull has 20 vertices, 37 edges and 19 facets, and volume 28, independently confirming the extrusion calculation. It also exhausts all 1,023 nonempty subsets of the ten cubes for an eligible planar subpolyomino. Numerical search was used only to find the example. No smallest-counterexample claim is made. This is an author-agent proof and audit, without independent peer review.

Proposed Tasks 2–5

2. Determine the smallest number of cubes in a three-dimensional counterexample to Kurz's recursive structure conjecture, with an exact classification or exclusion certificate for every smaller size.

3. Classify all ten-cube convex-hull maximizers up to translation, coordinate permutation and reflection, and determine which admit a subpolyomino satisfying the conjectured recursion.

4. Construct infinitely many sizes for which a three-dimensional convex-hull maximizer has no coordinate layer containing a connected subpolyomino with both full remaining side lengths.

5. Determine for each dimension $d \geq 4$ whether such a failure of the full-side-length layer condition occurs among convex-hull maximizers, giving constructions or a structural obstruction.

References

[1] S. Kurz, *Convex hulls of polyominoes*, Beiträge zur Algebra und Geometrie **49** (2008), no. 1, 125–136. Published article; arXiv:math/0702786v1 (2007).

[2] MathDB 380444, *Recursive structure conjecture for extremal polyominoes*. Source and public-claim audit dated 28 September 2026; details accompany the proof package.